계량경제학 강의 12

이분산

1. Gauss-Markov Theorem

$Y_i = \beta_1 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + e_i$

고전적 가정

(1) $E(e_i) = 0 \quad for\ all\ i$

(2) $E\left(e_i^2\right) = \sigma^2 \quad for\ all\ i$

(3) $E\left(e_i e_j\right) = 0 \quad for\ all\ i \neq j$

(4) $X : non - stochastic$

(5) $rank(X) = k : no\ exact\ multicollinearity$

1. Gauss-Markov Theorem

- Gauss-Markov Theorem : 고전적 가정이 성립할 때, 선형이고, 불편성을 가진 추정량중에 OLS가 BLUE 이다

$\hat{\beta}_{OLS}$, $\hat{\beta}_{other}$

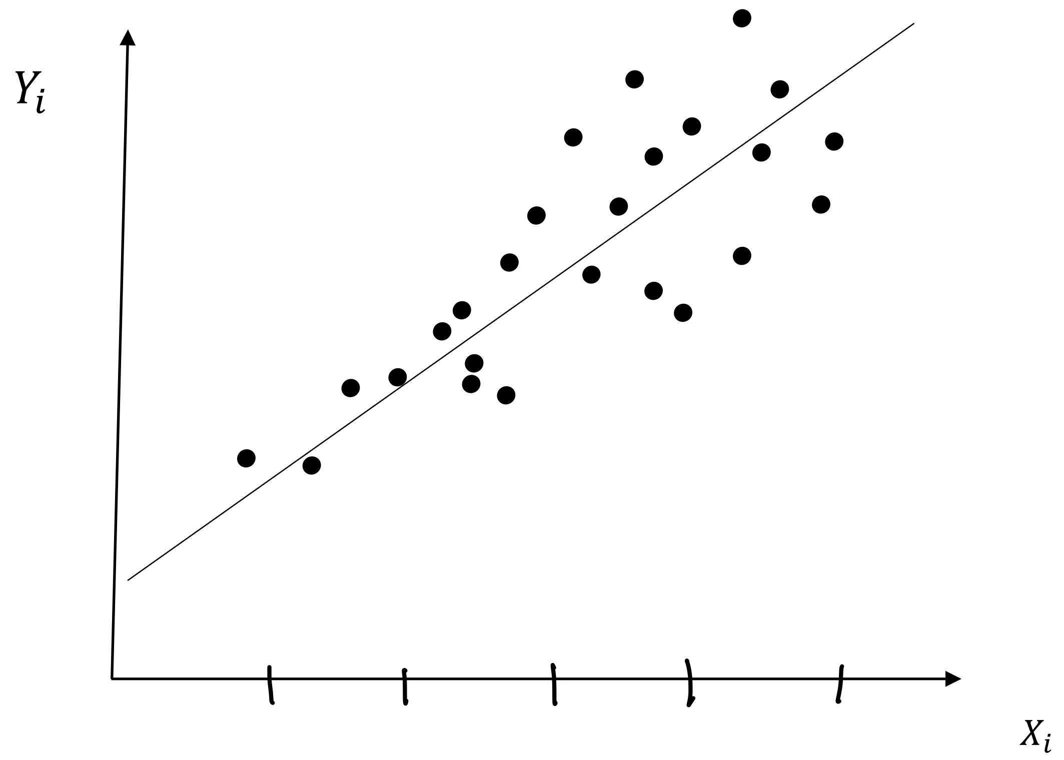
불편성 : $E(\hat{\beta}_{OLS}) = \beta$, $E(\hat{\beta}_{other}) = \beta$

선형 : $\hat{\beta}_{OLS} = \sum w_i y_i$ $\hat{\beta}_{other} = \sum w_i^* y_i$

$Var(\hat{\beta}_{OLS}) \leq Var(\hat{\beta}_{other})$

2. 이분산이 있는 단순회귀모형

예) 소득과 소비의 관계식 : $Y_i = \beta_1 + \beta_2 X_{2i} + e_i$



2. 이분산이 있는 단순회귀모형

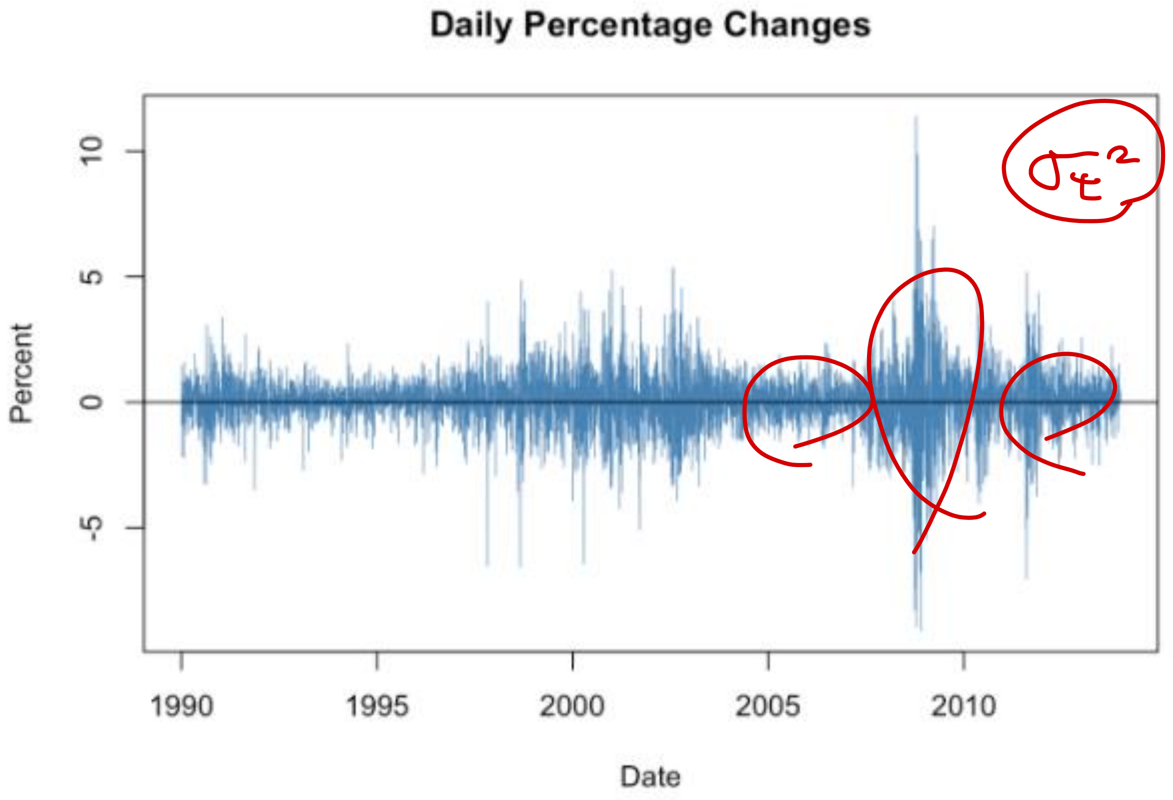


Figure 16.1: Daily Percentage Returns in the Wilshire 5000 Indrx

2. 이분산이 있는 단순회귀모형

≪ *Population Regression Equation* ≫

$$Y_i = \beta_2 X_{2i} + e_i,\quad e_i \sim (0, \sigma_i^2)$$

요인 X

σ^2

$$\widehat{\beta_2}_{OLS} = \frac{\Sigma X_{2i} Y_{2i}}{\Sigma X_{2i}^2}$$

$$E(\widehat{\beta_2}_{OLS}) = \beta_2$$

$Var(\hat{\beta}_{OLS}) \Rightarrow$ 大 최소 분산 X

추정의 정확도

정보의 효율성

분산 Big?

$$t\text{-value} = \frac{\hat{\beta}_2 - 0}{\sqrt{\hat{var}(\hat{\beta}_2)}}$$

BIG

$E(\hat{\beta}_{2,OLS}) = \beta_{2,OLS}$?

$$E\left(\hat{\beta}_2 = \frac{\Sigma X_{2i} \cdot Y_i}{\Sigma X_{2i}^2}\right) \quad Y_i = \beta_2 X_{2i} + e_i$$

$$\Rightarrow E\left(\frac{\Sigma X_{2i} \cdot (\beta_2 X_{2i} + e_i)}{\Sigma X_{2i}^2}\right)$$

$$= \beta_2 + \frac{\Sigma X_{2i} E(e_i)}{\Sigma X_{2i}^2} \quad (\because X_i : \text{non-stochastic})$$

$E(e_i) = 0$

$\therefore\ \hat{\beta}_{2,OLS}$ unbiased estimator

2. 이분산이 있는 단순회귀모형

OLS : not BLUE 2nd Best

$$Var(\widehat{\beta}_{2OLS}) = E\left[\left(\hat{\beta}_{2OLS} - E(\hat{\beta}_{2OLS})\right)^2\right]$$

by definition

$E(\hat{\beta}_{2OLS}) = \beta_2$

$$= E\left[\left(\hat{\beta}_{OLS} - \beta\right)^2\right]$$

$$\hat{\beta}_{2,OLS} = \frac{\Sigma X_{2i} Y_i}{\Sigma X_{2i}^2} = \beta_2 + \frac{\Sigma X_{2i} \cdot e_i}{\Sigma X_{2i}^2}$$

$$= E\left[\left(\frac{\Sigma X_{2i} e_i}{\Sigma X_{2i}^2}\right)^2\right]$$

$$= \left(\frac{1}{\Sigma X_{2i}^2}\right)^2 E\left(X_{21}^2 e_1^2 + X_{22}^2 e_2^2 + \cdots + X_{2n}^2 e_n^2 + 2 \cdot X_{21} \cdot X_{22} e_1 e_2 + \ldots\right)$$

$$= \left(\frac{1}{\sum x_{2i}^2}\right)^2 \Big[x_{21}^2 \underbrace{E(e_1^2)}_{=\sigma_1^2} + x_{22}^2 \underbrace{E(e_2^2)}_{\neq\ \sigma_2^2} + \cdots + x_{2n}^2 E(e_n^2) + 2\cdot x_{21}\cdot x_{22} \underbrace{E(e_1 e_2)}_{0} + \cdots \Big]$$

$$var(\hat{\beta}_{2.OLS}) = \left(\frac{1}{\sum x_{2i}^2}\right)^2 \sum_{i=1}^{n} x_{2i}^2 \sigma_i^2$$ — inefficient

If $e_i \sim N(0, \sigma^2)$,

$$var(\hat{\beta}_{2.OLS}) = \frac{\sigma^2}{\sum x_{2i}^2}$$

$$\boxed{\text{목표) 고전적 가정이 성립하도록 모형을 변환!}}$$

$\Downarrow$

Transformation 후, 그 모형은 "BLUE"

2. 이분산이 있는 단순회귀모형

이분산이 있는 모형을 고전적 가정이 성립하도록 바꿔보자

$\boxed{Y_i = \beta_2 X_{2i} + e_i,}$ $\boxed{e_i \sim (0, \sigma_i^2)}$

양변을 σ_i 로 나누면,

$\boxed{Y_i^* = \beta_2 X_{2i}^* + e_i^*}$, $e_i^* \sim (0, 1)$ $\boxed{\frac{e_i}{\sigma_i}} \sim (0, 1)$ $\Rightarrow e_i^*$

$$\underbrace{\frac{Y_i}{\sigma_i}}_{(= Y_i^*)} = \beta_2 \cdot \underbrace{\frac{X_{2i}}{\sigma_i}}_{= X_i^*} + \underbrace{\frac{e_i}{\sigma_i}}_{= e_i^*}$$

GLS 추정량 : $\widehat{\beta_2} = \frac{\sum X_{2i}^* Y_i^*}{\sum X_{2i}^{*2}}$

Generalized

2. 이분산이 있는 단순회귀모형

$E(\widehat{\beta_2}_{GLS}) = \beta_2$

$Var(\widehat{\beta_2}_{GLS}) = \dfrac{\sigma^2}{\sum X_i^{*2}}$ //

$by\ Gauss - Markov\ Theorem, Var(\widehat{\beta_2}_{OLS}) > Var(\widehat{\beta_2}_{GLS})$

(고전적가정이 성립했을때 OLS : BLUE)

2. 이분산이 있는 단순회귀모형

$\sigma_i \longrightarrow \hat{\sigma}_i$

- 이분산의 패턴을 모른다면, 여전히 OLS 추정량을 이용해야 한다.
- 다만, 가설검정을 할 때 제대로 된 분산을 이용해야 한다

GLS) σ_i : known

현실) σ_i : X unknown,

$\downarrow$

$\hat{\sigma}_i \Rightarrow$ FGLS

(Feasible GLS)

3. 이분산이 있는 다중회귀모형

$\ll$ *Population Regression Equation* $\gg$

$Y = X\beta + e, \qquad \underset{n\times 1}{e} \sim (0, \underset{n\times n}{\sigma^2\Omega})$

$\underline{Var(e)} = \underset{n\times n}{\underline{E(ee')}}$

$= \sigma^2\Omega \neq \sigma^2 I_n$

$$\begin{bmatrix} var(e_1) & cov(e_1,e_2) & \cdots & cov(e_1,e_n) \\ cov(e_2,e_1) & var(e_2) & & \\ & & \ddots & \end{bmatrix}$$

① $\sigma^2\begin{bmatrix} k_1 & & & 0 \\ & k_2 & & \\ & & \ddots & \\ 0 & & & k_n \end{bmatrix}$ (이분산)

② $\sigma^2\begin{bmatrix} 1 & & & \text{non-zero} \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$ (자기상관모형)

③ $\sigma^2\begin{bmatrix} b_1 & & & \text{non-zero} \\ & b_2 & & \\ & & \ddots & \\ \text{non-zero} & & & b_n \end{bmatrix}$ (이분산 + 자기상관)

Q) $\sigma^2\Omega$에서 이를 무시한 OLS?

$\widehat{\beta}_{2OLS} = (X'X)^{-1}X'Y$

$= (X'X)^{-1}X'(X\beta + e) = \beta + (X'X)^{-1}X'e$

$E(\widehat{\beta}_{2OLS}) = \beta + (X'X)^{-1}X'\underset{=0}{E(e)}$

$= \beta$

3. 이분산이 있는 다중회귀모형

$(AB)' = B'A'$

by definition

$$Var(\widehat{\beta}_{2OLS}) = E\left[\left(\hat{\beta}_{OLS} - \underset{=\beta}{E(\hat{\beta}_{OLS})}\right)\left(\hat{\beta}_{OLS} - \underset{=\beta}{E(\hat{\beta}_{OLS})}\right)'\right]$$

$$\hat{\beta}_{OLS} - E(\hat{\beta}_{OLS}) = (X'X)^{-1}X'e$$

$$= E\left[(X'X)^{-1}X'e \; e'X(X'X)^{-1}\right]$$

$$= (X'X)^{-1} X' \boxed{E(ee')} X (X'X)^{-1}$$

$$E(ee') = \sigma^2\Omega$$

$$= \underset{잼}{\sigma^2} \underbrace{(X'X)^{-1}}_{빵} \underbrace{X'}_{고기} \underset{토마토}{\Omega} \underset{고기}{X} \underbrace{(X'X)^{-1}}_{빵} \; : \text{Sandwich formula.}$$

<OLS>

① unbiasedness ok!

② 다만, OLS - not BLUE

(not min. variance)

③ Eviews, Stata, python, default option, 이들 무시하고 output.

제대로 된 분산은

$$var(\hat{\beta}_{OLS}) = \sigma^2 (X'X)^{-1} X' \Omega X (X'X)^{-1}$$ 인데,

컴퓨터는 $var(\hat{\beta}_{OLS}) = \sigma^2 (X'X)^{-1}$ 를 사용.

$$t\text{-value} = \frac{\hat{\beta}_2 - 0}{\sqrt{\widehat{var}(\hat{\beta}_2)}}$$ 모두 wrong!

3. 이분산이 있는 다중회귀모형

$Y = X\beta + e, \quad e \sim (0, \sigma^2 \Omega)$ $n \times n$

- GLS 추정량

촐레스키 분해(*Cholesky Decomposition*)

$C: 3\times 3$

: $\boxed{\Omega^{-1}} = C'C$ C : Lower triangular matrix ⇒ $\begin{bmatrix} * & & \\ & \ddots & \\ 0 & 0 & \end{bmatrix}$

positive definite : 고유값이 양수인 행렬

모집단회귀식 양변에 C를 곱해주면,

예)
$\pi = f_1(i, Y, E\pi)$
$i = f_2(i, Y, E\pi)$
$Y = f_3(i, Y, E\pi)$

$Y = X\beta + e$, $\boxed{e \sim N(0, \sigma^2 \Omega)}$

$(\Omega^{-1} = C'C)$

$\underbrace{C \cdot Y}_{Y^*} = \underbrace{C \cdot X}_{X^*}\beta + \underbrace{C \cdot e}_{e^*}$

$e^* \sim N(0, \sigma^2 C \cdot \Omega C')$

$= C \cdot e^* \sim N(0, \sigma^2 I_n)$

3. 이분산이 있는 다중회귀모형

$$Y^* = X^*\beta + e^*, \qquad e\sim(0,\sigma^2 I_n)$$

$$\widehat{\beta}_{2GLS} = \left(X^{*\prime} X^*\right)^{-1}\left(X^{*\prime} Y^*\right) = (X'\underbrace{C'C}_{\Omega^{-1}}X)^{-1}(X'\underbrace{C'C}_{\Omega^{-1}}Y)$$

$$= (X'\boxed{\Omega^{-1}}X)^{-1}(X'\boxed{\Omega^{-1}}Y)$$

$$E(\widehat{\beta}_{2GLS}) = \beta_2$$

$$Var(\widehat{\beta}_{2GLS}) =$$

$\boxed{1}$ $\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}\cdot Y$

$E(\hat{\beta}_{GLS}) \stackrel{?}{=} \beta$

$= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}(X\beta + e)$

$E(\hat{\beta}_{GLS}) = \beta + (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}e$ —— ?

$\cancel{E(e)}^{\,0}$

$\therefore E(\hat{\beta}_{GLS}) = \beta.$

$\boxed{2}$ $Var(\hat{\beta}_{GLS})$ $\Downarrow$

$= E[(\hat{\beta}_{GLS} - \beta)(\hat{\beta}_{GLS} - \beta)']$

$= E[(X'\Omega^{-1}X)^{-1}X'\cancel{\Omega^{-1}}\underbrace{ee'}_{\Downarrow\ \sigma^2\Omega}\cancel{\Omega^{-1}}X(X'\Omega^{-1}X)^{-1\prime}]$

$= \sigma^2(X'\Omega^{-1}X)^{-1}$ //

<Summary>

$Y = X\beta + e, \quad e \sim (0, \sigma^2\Omega)$

OLS	GLS
$\hat{\beta}_{OLS} = \boxed{(X'X)^{-1}X'Y}$	$\hat{\beta}_{GLS} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}Y$
$E(\hat{\beta}_{OLS}) = \beta$	"
$Var(\hat{\beta}_{OLS}) = \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}$	$Var(\hat{\beta}_{GLS}) = \sigma^2(X'\hat{\Omega}^{-1}X)^{-1}$
A	B

$\hat{\square}$: FGLS

BLUE

(Best linear Unbiased Estimator)

$\underline{\underline{Var(A)}} - \underline{\underline{Var(B)}} \geq 0$

Ω : GLS

Ω unknown, able to estimate

: FGLS

Ω unknown. 추정불가 : OLS.

______ _ (hetero)